title	authors	year	doi	Map representation	Map matching	Dynamic objects detection	Prediction modeling	
Probabilistic Object Maps for Long-Term Robot Localization	Adkins, Amanda Chen, Taijing Biswas, Joydeep	2022	10.1109/IROS47612.2022.9981316	-	-	-	Probabillistic Object Maps (POM) to model the likelihood of a movable object of a given semantic class ocurring at a given pose: use a variante of Gaussian Process Classification (GPC) to estimate the likelihood. Compute the covariance of the estimation using an approach derived from desired properties of variance: more sample inputs should result in lower variance + samples closer to query pose should result in lower variance than same number of distance samples >>> reciprocal of an unnormalized kernel density estimator (KDE). POM-Localization incorporates odometry and object detections to estimate belief over the robot's trajectory, introducing a POM observation likelihood general to several semantic forms of object detections. POM-Localization significantly outperforms raw wheel odometry, cartographer, and LeGO-LOAM. EnML (Episodic non-Markov localization, Biswas and Veloso) has slightly better average-case performance, however POM-localization has better or comparable worst-case performance.	
Exploring the Effect of 3D Object Removal using Deep Learning for LiDAR-based Mapping and Long-term Vehicular Localization	Chen, Yifan Sun, Shuo Yin, Huan Ang, Marcelo H.	2022	10.1109/ITSC55140.2022.9921969	-	-	Dynamic object removal module using PointPillars and a localization and mapping module based on LOAM (removal based on class label!). 1. PointPillars selected as main 3D object detection network given its high accuract + ability process point clouds in real time 2. removal of the points that fall within the detection bounding boxes, considered as outlier points 3. processed point cloud used for localization and mapping	-	
Learning Object-Based State Estimators for Household Robots	Du, Yilun Lozano-Pérez, Tomás Kaelbling, Leslie Pack	2022	10.1109/IROS47612.2022.9981287	-	-	-	Develop method for training neurl networks to perform as Object-Based Memory (OBM) for dynamic entity monitoring, training system to construct memory of objects in the environment wo/ explicit models of sensors, types of objects, nor patterns in which they might occur. Designed the OBM-Net architecture to learn OBMs: 1. network structured as recursive filter, with new memory values computed on each observation and fed back to the newxt 2. order in which object hypotheses are delivered in should not matter + memory structure should also be permutation invariant and independent of number of objects >>> memory processing as attention mechanism 3. memory operate on latent representation related to observations and output hypotheses via encoder and decoder modules, given observations may not be well syted for hypothesis representation Experiments evaluate OBM-Net on several different entity monitoring tasks: goal is for the robot to be able to construct hypotheses for each distinct object it has seen and to be able to predict for each object the table it is currently on and its location relative to the table. Compare to STRANDS Project (complex spatial-temporal clustering method >>>CHECK IT OUT IF IT IS KRAJNIK) + DS-SLAM (Yu et al. IROS 2018). OBM-Net outperform other methods substrantially better at long-term memory. OBM-Net learns from experience how to perform as a data-association filter in novel environments with complex observation and transition models that would be too difficult to hand-specify. Importantly, our system may be applied	
LT-mapper: A Modular Framework for LiDAR-based Lifelong Mapping	Kim, Giseop Kim, Ayoung	2022	10.1109/ICRA46639.2022.9811916	Change composition, where once change is detected, decision for map maintenance determines what to include or exclude. Method also extracts stable structures with higher placeness, constructing a reliable 3D map with authentically meaaningful structures for other missions >>> LT-map to manage changes and enable central map to evolve in place-wise manner. LT-map can produce live map (up-to-date) vs persistent map (meta map) efficiently, while storing changes as delta map, where by exploiting delta maps, restoration and change detection become memory and computation cost-effective! LT-map: negative difference map elimination + positive difference map merging LiDAR-based multi-session 3D SLAM module - LT-SLAM!	-	Scan-to-map scheme with projective visibility, using range images of multiple window sizes LT-removert - extend Kim&Kim IROS 2018 to include both intra/inter session change detection, segretating high and low dynamic (HD and LD, respectively) objects from change (Wallcott-bryant et all iros2012). Use Kim&Kim IROS'18 HD pts removal, using image-based discrepancy and different sizes of windows to alleviate the pose ambiguity Once two sessions are aligned and HD pts within them were removed, compare query and central sessions to parse LD pts, constructin KDtree for target map and test whether source map's point has k target pts within a threshod (if not, the point is classified LD) >> then parse positive difference (PD) and negative difference (ND) of added v s removed pts, respectively	-	
Feature Based Multi-Hypothesis Map Representation for Localization in Non-Static Environments	Nielsen, Kristin Hendeby, Gustaf	2022	10.23919/FUSION49751.2022.9841255					
Efficient 2D LIDAR-Based Map Updating For Long-Term Operations in Dynamic Environments	Stefanini, Elisa Ciancolini, Enrico Settimi, Alessandro Pallottino, Lucia	2022	10.1109/IROS47612.2022.9982047	input: occupancy grid map build with any SLAM algorithm --> build geometrically and temporally representation of the environment suitable for any continued localisation algorithm. Cell represent state that corresponds to cell probab free (0) occupied (1), or unknown (otherwise) --> occupancy probab updated from 2D laser scan measurements.	-	1. beams classifier classify them as "detected change measurement" 2. "changed + unchanged cells evaluator" evalute cells associated to measurements w.r.t. those in the initial map to confirm the type of detection 3. do not change the cells based on a single measurement: for each cell, rolling buffer of fixed dimension, filled with outcomes of evaluator blocks at different time instants - eucliden distance between measured hit point and expected one, comparison between one measurement vs N set of points in map Evo Python packate to analyse localisation errors with and without updated maps in simulation experiments: AMCL pose based on initial map vs last available update >>> using updated map reduces greatly errors, while using the outdated map robot bumps into some objects (aaa well quite a stupid robot!) No conclusions made on localisation in real experiments due to not having ground-truth...	-	
Dynamic Visual SLAM Based on Efficient Two-Stage Statistical Model	Wu, Shaoteng Zhang, Ruitao Wang, Yafei	2022	10.1109/CVCI56766.2022.9964605	-	-	efficient two-stage statistical model for dynamic landmark removal: static scores of landmarks estimated to obtain initial static/dyamic labels and a conditional random field constructured to refine the labels. 1. extract FAST keypoints and ORB descriptors similar to ORB-SLAM2 2. during tracking of visual odometry, static score specific landmark estimated by statistical characteristic of re-projection error (Euclidean distance?) and then propagated among subsequent frames 3. conditional random field in consideration of neighbor motion and historical observation built on fully connected graph of all landmarks to improve spatiotemporal consistency of static/dynamic labels - some landmarks belonging to same moving object may have static labels as consequence incorrect data association or relatively slight observed motion from camera view 4. ultimately dynaic label eliminated while static ones applied to state estimation compared accuracy and efficiency base ORB-SLAM2 + other methods (including n LC-CRF-SLAM!!!) to proposed approach - pwithout second stage labelling, method significantly promotes ORB-SLAM2 performances - full proposed method improve ATE and RPE by 80-90% - although LC-CRF-SLAM shows lowest RMSE of ATE and RPE on all sequences, accuracy proposed method is at same level with only few centimeters disadvantage - comparing with LC-CRF-SLAM on same hardware, results show that significant efficiency advantage over LC-CRF-SLAM (in computation time)	-	
DE-SLAM: SLAM for highly dynamic environment	Xing, Zhiwei Zhu, Xiaorui Dong, Dingcheng	2022	10.1002/rob.22062					
Long-Term Visual Simultaneous Localization and Mapping: Using a Bayesian Persistence Filter-Based Global Map Prediction	Deng, Tianchen Xie, Hongle Wang, Jingchuan Chen, Weidong	2023	10.1109/MRA.2022.3228492	time-series global map built to save previous observations and obtain temporal maps of environment, where each map consists of keyframes and map points, where each map points stores: - 3D position - vieweing direction - survival-time - persistence probability of map pt (output of persistence filter) >> used to classify map points into static, semistatic, and dynamic - representative map point descriptor (different from original ORB) - maximum and minimum observation distance, according to sensor characteristics	-	offline time-series modelling: - map points matching: used to obtain accurate time series of map points - Bayesian Persistance Filter (BPF): knwoing that each map point has its survival time (0,+inf.), persistance variable is a boolean random variable representing whether map point exists or not at a time t - semistatic map points cluster modules Global Map Prediction (GMP): predict future state of semistatic map poitns and obtain predicted global map, persistence of map point clusters calculated, and predicted maps constructed with long shot-term exponential model prediction persistence filter evalutes persistence of map points and removes dynamic points TUM dataset evaluations compared to ORB-SLAM2, ORB-SLAM 3, DynaSLAM (Bescos RAL'18) - method more accurate than ORB-SLAM2 and ORB-SLAM3 - while methods like DynaSLAM that are learning-based ones mask the entire image region of a person, most body is static in low-dynamic environments, resulting in less environment information used and #available static points for pose estimation not enough resulting in poor localization performance - however in high dynamic environments, proposed method not good enough given they are "supposedly" designed for dynamic TUM dataset	-	
Multi-Hypothesis SLAM for Non-Static Environments With Reoccurring Landmarks	Nielsen, Kristin Hendeby, Gustaf	2023	10.1109/TIV.2022.3214978	extends FUSION'22 work by considering creation and updating feature-based multi-hypothesis map representation: - theoretical incorporation of feature-based multi-hypothesis map representation into SLAM framework, still allowing underlying static SLAM problem be solved by any state-of-the-art SLAM solver of choice - tree structure for efficient hypothesis management and decision making is introduced - formalization of multi-hypothesis SLAM algorithm, reinforced with ingredients from target tracking theory, allowing missed detections and false alarm observations to be modeled Hypothesis tree: each node represents one hypothesis associated w/ prior probability and a recursively updated score >>> tree can be dynamically updated as the robot is moving and nodes can be added when new hypothesis evaluation is instantiated - tree growth typically triggered when observation of landmark not gated w/ any known position, landmark w/ knwon multiple nodes enters FOV, or missed detection noticed - nodes removed when decisions are made or hypotheses merged Paper considers scenario where uniquely identifiable landmarks present and can be moved between multiple discrete positions, not known a priori				
Visual Localization and Mapping in Dynamic and Changing Environments	Soares, Joao C. Medeiros, Vivian S. Abati, Gabriel F. Becker, Marcelo Caurin, Glauco Augusto de Paula Gattass, Márcelo Meggiolaro, M.A.	2023	10.1007/s10846-023-02019-6	-	-	Changing-SLAM (proposed method) based on ORB-SLAM3: - robustness to highly dynamic environments (keypoint classification that filters a priori dynamic objs + EKF track movable objects in scene) - robustness to changing environments (use 3D mapped pts derived from feature detection + output of object detector to determinethe 3D centroid of objs in the scene) - works by giving semantic and temporal meaning to mapped pts: pre-vilter of dynamic pts to prevent odometry drift vs map management system prevent wrong loop closures Object belief map created and updated, making SLAM process occur in two levels: within high level (objs) Bayesian filter is used to create belief map about poses of objects in the map (belief map used to determine which features used for low-level step); within low level (pts) SLAM problem solved using feature-based methods proposed by ORB-SLAM3, adapted for dynamic environments Keypts initially classified into three categories: dynamic, movable, static. All keypts belonging to people are classified as a priori dynamic, keypts belong to objs classifid as movable, and keypts from background classified as static. - 2 keypts that belong to same obj CANNOT have different classification - using object detection for feature rmv can lead to feature depletion >> propose efficient method to correctly classify features that belong to bounding box but are not dynamic, called feature repopulation, based on the median+mean+max+min px depth + std deviation + intersection over union matrix to evaluate wether detected obj being occluded and to differentiate the object from background if keypts classified the object as static, keypts used for feature tracking Experiments compare proposed method with DynaSLAM, DS-SLAM. Overall TUM dataset DynaSLAM and SaD-SLAM had best results but methods are not realtime (300ms per frame wo/ considering time for instance segmentation for DynaSLAM, SaD-SLAM only offline), while Changing-SLAM achieved average tracking speed of 23.8fps (well is the hardware the same you buggars?).	-	
Safe and Robust Map Updating for Long-Term Operations in Dynamic Environments †	Stefanini, Elisa Ciancolini, Enrico Settimi, Alessandro Pallottino, Lucia	2023	10.3390/s23136066	previous method relies heavily on accurate knowledge about initial pose of robot in previously built map--> integrate recovery procedures to ensure map accuracy not degraded during long-term operations + fail-safe mechanism added based on localisation performance to prevent erroneous map updating when there is an increase in localisation error + integration of pose update activeted when robot effectively gets lost large localization error may lead to an incorrect update of the map --> hence the localisation check! If error becomes too large, pose update requested to dedicated algorithm				
Continual SLAM: Beyond Lifelong Simultaneous Localization and Mapping Through Continual Learning	Vodisch, Niclas Cattaneo, Daniele Burgard, Wolfram Valada, Abhinav	2023	10.1007/978-3-031-25555-7_3	Continual (Continuas Learning) SLAM (CL-SLAM): dual-network architecture of visual odometry (VO) model consisting of an expert that produces myopic online odometry estimates + generalizer focuses on long-term learning across environments. Training both networks in self-supervised manner where weights of expert updated only based on online data + weights of generalizer updated based on combination of data from both online stream + replay buffer. VO estimates of expert construct pose graph. Monodepth2 jointly predicts depth map of an image and the camera motion from previous timestep to the current. Loop closure detection performing place recognition using pre-trained and frozen CNN LoopNet: 1. map every frame to feature vector using MobileNetV3 small trained on ImageNet + store them in dedicated memory 2. compute cosine similarity of current feature map with all preceding feature maps: if sim_cos above given threshold, use PoseNet compute transformation between corresponding images Dynamic objects detection: Mask all potentially dynamic objects using bounding boxes generated by YOLOv5m trained on the COCO dataset. CL-SLAM outperforms DeepSLAM on majority sequences on KITTI data even though it waas only exposed to Cityscapes dataset before deployment. Compared to ORB-SLAM, CL-SLAM suffers from higher rotation error but can improve the translation error in sequences 6 and 10 of KITTI dataset				Additional obs: - Adaptation Quality (AQ): measures short-term adaptation capability when being deployed in new environment - Retention Quality (RQ): captures long-term memory retention when revisiting previously encountered performances
BioSLAM: A Bioinspired Lifelong Memory System for General Place Recognition	Yin, Peng and Abuduweili, Abulikemu and Zhao, Shiqi and Xu, Lingyun and Liu, Changliu and Scherer, Sebastian	2023	10.1109/TRO.2023.3306615	BioSLAM has 2 memory zones: static memory zones (SMZs) for historical memory encoding w/ low frequency and dynamic memory zone (DMZ) for quickly memory reply + dual-memory selection mechanism to balance short-term adaptation for new observations and long-term memory retention for historical knowledge. BioSLAM develops sleeping cycle for memory consolidation within SMZ, inspired bya similar mechanism in the human hippocampus, achieving long-term Place Recognition. BioSLAM as an incremental PR method that enable continual learning of place feature extraction modules across various domains: 1. general place feature extraction module for ecoding place features; 2. BiLM (bioinspiered lifelong memory) for continual learning PR for different domains. - rewarding mechanism to measure memory codes' importance / reward calculation -> importance score will be used to manage memory consolidation and selection in dual-memory module - dual-memory module to cooperate w/ rewarding mechanism for long-/short-term memory storage and importance-retrieval w/ limited space usage SMZ consolidation:feature-spatial code for memory trace to capture both spatial and feature properties -> K-means clustering, where memory traces within single cluste have similar characteristics (redundant data) -> downsmapling within each cluster. SMZ forgetting: eliminates redundant memory clusters that are too similar. DMZ refreshing: given DMZ bref and storage limited properties, clone relevant memory traces from SMZ to DMZ (traces with higher rewards assigned higher sampling weights as they have lower localization ability and robustness, requiring more attention from BioSLAM + decaying mechanisms to encourage DMZ to be more curious about new traces. DMZ replay: replay selected memory traces to memory decoder for generating synthetic sampes. **System showed stron adaptability to changes in domains with its performance not dropping significantly when shifting from one domain to another, e.g., LiDAR, daytime, and night-time visual domains!**	-	-	-	
Lifelong SLAM Based on Environmental Semantic and Depth Information in Dynamic Scenes	Yu, Naigong and Wang, Yang and Gan, Mengzhe and Ma, Nan and Huang, Gao	2023	10.1109/ACAIT60137.2023.10528652	-	-	(1) lightweight semantic segmentation network is designed to segment dynamic target in the tracking thread, generating semantic information for locating dynamic objects. Mask used to extract static feature points in the image, in order to improve localization accuracy and robustness of system in dynamic scenes. (2) in loop closing thread, add depth information constraints to visual features that may move in scene. (3) new static map keyframe selection strategy to constructu static semantic map. MobileNetV2 replaces ask R-CNN + front-end network also pruned for real-time performance purposes: 1. network first extracts features based on MobileNetV2; 2. inputs channel feature map into ROI polling layer; 3. give object classification and position by classifier and non-maximum supperssion Focus on data classes containing dynamic objects such as "humans", "animals", and "cars" for training: make semantic annotations of specific types of objects and remove dynamic visual features from the image belonging to the respective dynamic objects in the scene. (1) semantic annotation of keyframes (2) ORB feature extraction at the same time + dynamic region segmentation performed by combining keyframe semantic information (3) perform visual word statistics on current frame to obtain BoW vector (4) filter keyframes according to number of shared workds + calculate loop candidate keyframes by similarity of features (5) compare depth of current frame features with map points + loop detection through depth constraint of the loop candidate keyframe **Paper adds 3D static semantic map building module to ORB-SLAM2 framework** Compared to ORB-SLAM2 on TUM daataset, accuracy of proposed system significantly improved and RMSE improvement reaches 95% on average.	-	
Generalizable Stable Points Segmentation for 3D LiDAR Scan-to-Map Long-Term Localization	Hroob, Ibrahim and Mersch, Benedikt and Stachniss, Cyrill L. and Hanheide, Marc	2024	10.1109/LRA.2024.3368236	-	-	Use sparse convolutions due to their efficiency to segment scaan point cloud data into to caategories: stable and unstable pts. Proposed approach considers stable pts segmentation that data can be segmented both present dynamic entities and stationary objects that may undergo popsitional or perceptual alterations in subsequent instances. Also, method can be trained in self-supervised manner with no manual annotation by leeraging previous observations! Goal: scan-to-map matching in changing environments, training4D sparse convolutional neural network in self-supervised manner, predicting spario-temporal features in current scan through exploitation of discrepancies between scan and map data. Use stable pts to increase accuracy of robot long-term localization + generalize across diverse and unseen environments, while suitable for online operation on mobile robot. Proposed approach was able to segment unstable pts across novel and unseen environments. Improves results on LTS-NETin terms of generalization, given that LTS-NET segments based on object shapes seen during training.	-	
Adaptive robot localization in dynamic environments through self-learnt long-term 3D stable points segmentation	Hroob, Ibrahim and Molina, Sergi and Polvara, Riccardo and Cielniak, Grzegorz and Hanheide, Marc	2024	10.1016/j.robot.2024.104786			Long-Term Stability NETwork (LTS-NET): regression network capable of learning spatiotemporal labels from auto labelling algorithm directly on point cloud data, using PointNet++ architecture to process point cloud directly. - LTS-NET encoder aggregate local features from previous laery into global feature representation for each point set + decoder to restore point features from down-sampled point cloud to its original form - use NDT algorithm as matching algorithm Filtering scans based on long-term stable features significantly enhances localization performance and robustnees in dynamic environments., while performing well in entirely new environments.		

Real-Time Submap Trimming-Based Map Updating for Long-Term Localization of Mobile Robot in Dynamic Environments	Li, Shuaiyong and Li, Tengfei	2024	10.1109/TII.2024.3413958	-	Real-Time Submap Trimming (RTST), based on cartographer!: - (cartographer) loop-closure constraints established between new laser scanning frame nodes and historical submaps through global map scan matching - RTST module dynamically streamlines global map data by identifying and trimming redundant submaps in real time (submap centroid fefature extraction, submap overlap rate recognition, submap observation gain model, trimming of redundant submaps) Submap centroid feature used in submap overlap rate recognition part, where high overlap rate suggests that submap significantly overlaps with other submaps, containing numerous duplicated environmental features, can be considered redundant Submap observation gain model computes number of constraints from pose graph and corresponding constraints for each submap, where these values used to determine gain value of each submap, reflecting validity of submap's observation information, storing outdated information can cause buildup of irrelevant data + negatively affect accuracy of localization in long run. "occuapncy probability is lower when dynamic objects change rapidly and higher when environmental features are stable during craetion of new submap" Evaluated in lifelong SLAM dataset, collected in Beijing, China from 28/09 and 31/10, 2020. Ground-truth data using Adaptive Monte Carlo Localization (AMCL). Proposed method outperformed odom only and localization without map updates, significantly improving localization accuracy by anaverage of 34.83% on ATE.	-		possible interesting lifelong dataset for 2D data: A General Framework for Lifelong Localization and Mapping in Changing Environment. IROS 2021 (https://github.com/sanduan168/lifelong-SLAM-dataset)
CTO-SLAM: Contour Tracking for Object-Level Robust 4D SLAM	Li, Xiaohan and Liu, Dong and Wu, Jun	2024	10.1609/aaai.v38i9.28899	-	-	CTO-SLAM (Contour Tracking for Object-level robust 4D SLAM - estimate camera motion + dynamic object motion simultaneously - dynamic object motion initialization algorithm to generate poses + contour tracking to track contour keypts w/ pixel tracking - goal: accuratly reconstruct static and dynamic elements of environment (1) instance-level semantic segmentation to segment and identify potentially movable objs within scene, by aligning masks with Intersection over Union (IoU) to determine which instance represents truly dynamic obj (2) system built upon ORB-SLAM 2 (2017) for camera pose estimation Object motion tracking: 1. dynamic pose initialization: estimate 6DoF pode when first seen 2. object contour tracking: track countour keypoints, extracted from semantic information in the edge of dynamic objects + pixel tracking establishes robust and effective spatiotemporal association in 2D image plane over long period 3. dynamic pose estimation: novel contour tracking based bundle adjuustment designed to optimiize pose from above two modules within sliding window All countour keypts, camera poses, object poses, annd sampling-based poses integrated to form factor graph and optimize within it + increase weight of sampling-based poses in factor graph Incorporate unique method for tracking object contours and optimizing among cameras, objects, and 3D pts., being able to establish robust long-term spatiotemporal associations; tightly integrates cameras, objects ,static keypts and countour keypts --> experimental results indicate that as static structure is enriched, accuracy of camera motion estimation improves significantly + CTO-SLAM showed 20% accuractimprovement over DynaSLAM, DynaSLAM2, and ORB-SLAM2 in RPE, while 55% on ATE over ORB-SLAM2 and 28% DynaSLAM and 32% on DynaSLAM2 respectively on KITTI Tracking datasets to evaluate ego-motion estimation	-	
Object-Oriented Grid Mapping in Dynamic Environments	Pekkanen, Matti and Verdoja, Francesco and Kyrki, Ville	2024	10.1109/MFI62651.2024.10705762	NDT-OM: divides environment into regular 3D grid of cells, which each represents 3D normal distribution, number of measurements, and probability of being occupied C-NDT-OM extends previous by storing cluster index, cluster membership, and categorical distribution of semantic labels	-	Propose way to use all measurements to update cells in grid map by jointly estimating posterior occupancy of map + set of latent variables that describe correspondence between cells occupied by same object! --> by aggregating evidence from cells occupied by same obj, map should adjust faster to environment's dynamics + allows infer occupancy of occluded cells tbelonging to partially occluded object reducing number of residual cells left by occlusions, improving map quality **(world consists of objects that span multiple cells)** Propose extension to Normal Distributions Transform Occupancy Map (NDT-OM) called Clustered-NDT-OM (C-NDT-OM) which uses proposed method + estimates latent variables by clustering cells based on their semantic labels lattent variable that models effect on cell mi of the measurement zj i.e. those falling on cell mj; in order to reflect that if there is a positive latent variable between two cells, reflects that the same ocject occupies both cells Usage of region growing algorithm to implement clustering on latent variables Proposed method outperformed NDT-OM in highly dynamic environments, resulting in fewer redisual cells from dynamic objs. Still, proposed method one of most common failures is erronuous joining of disjoint clusters of same label due to their proximity that may lead to the false removal of semi-static structures.	-	
Long-term navigation for autonomous robots based on spatio-temporal mapprediction ☆	Wang, Yanbo and Fan, Yaxian and Wang, Jingchuan and Chen, Weidong	2024	10.1016/j.robot.2024.104724	map database icluding semi-static maps + dynamic density maps through different spatial descriptions. Use HectorSLAM as basis for the mapping process (simplicity and efficiency?). - semi-static maps: contain static + semi-static objects derived directly from raw map obtained frfom original HectorSLAM (1. remove static objects for semi-static object clustering; 2. overlay semi-static object maps of all moments together + K-means get label for each grid and generate clustered semi-static object map) - dynamic density maps: goal is for autonomous robot automatically select path with fewer highly dynamic objects in advance from global path planning level through long-term map prediction	-	-	Autoregressive Moving Average map prediction based navigation (Arm_MPNAV) 1. unified spatio-temporal map prediction framework for objects with different degrees of dynamics, using AutoRegressive Moving Average (ARMA) model to describe ad predict changing patterns for both semi-static and highly dynamic objs 2. novel global path planning algorithm for long-term navigation, based on predicted map and interpolation strategy, build time-variante predicted cost maps + time-variant A* algorithm (TVA*) to acquire global path. Build several ARMA models (Hassler ARMA Stochastic processes calculus springer 2016) for semi-static maps and dynamic density maps through model order estimation, model parameter calculation, and constant model update --> define dynamic object density to reflect possibility that dynamic objects appear in certain cell - #times cell is occupied / (area of the cell x time interval of data collected by laser) ARMA used to identify time-variant patterns from data, where ARMA describes stationary random process by polynomials and analyzes given time series data, deals with both periodic and non periodic regular changes. Map prediction: predict state of clusters and dynamic object desnsity. Map update: continuously update current observations to predicted semi-static map + correct prediction errors to obtain an updated map. Long-term path planning generates feasible paths with low localization uncertainty and fewer dynamic objects. Evalute performance on spatio-temporal map prediction, comparing Arm_MPNAV and FreMEn on simulated large-scale parking lot with periodic and aperiodic changing car patterns. Since FreMEn is based on spectral model, cannot effectively handle non-periodic growth pattern. Also evalute real parking lot with the JiaoLong smarl wheelchair. Arm_MPNAV more accurate than FreMEnn for most parking spaces, being more stable for various changing patterns in real world.	
Lifelong 3D Mapping Framework for Hand-Held & Robot-Mounted LiDAR Mapping Systems	Yang, Liudi and Prakhya, Sai Manoj and Zhu, Senhua and Liu, Ziyuan	2024	10.1109/LRA.2024.3417113	map version control: - reconstruct previous session maps, performing backward progression of map version control, only require current base map + saved positive & negative differences + corresponding boundaries of each session map --> use HullFilter from PCL library that takes in stored session map boundary pts at time k and operate/crop base map to reproduce desired session map - forward progression computes based map from coexist + base overlap + base nonoverlap + session nonoverlap + session positive diffence - base negative difference	Map change detection identifies the changes between base map and incoming session map --> process them by a k-NN-based spatial filter to create base diff + coexist + session diff, where radius search performed for each base map pt in session map, classifying pts with radial neighbors as coexist and those wo/ as base diff + repease same process for each pt in session map, searching in base map --> another spatial search for each base diff pt in coexist classifying pts with radial neighbor as base overlap and those wo/ as base nonoverlap while repeating the same process for session diff and coexist pts. --> BEV descriptor compress spatial info of 3D point cloud into 2D matrices to accelerate process of comparison for map change detection,	process input LiDAR scans + poses using OctoMap to generate static ma, dynamic map, and set of unknown pts (may have residual holes in planar surfaces due to imprecise probabilities at voxel boundaris + created static maps much sparser and contain residual dynamic pts) -->> create submap concatenating 10-50 frames > RANSAC plane regression >> larger plane added to static map from OctoMap (ground / large building facade) > ratio #pts plane / #submap pts to determine if large enough and added to static map, filling residual holes + optional height filter that classifies all pts above certain height as static. K-NN voting to categorize unknow pts from OctoMap into static / dynamic --> statistical outlier removal + perform radial filtering to reassign pts close to static pts in dynamic map as static. Experiments with datasets (KITTI, NCLT, LT-ParkingLot, MulRan, etc.) proposed approach precisely detects walking human + dynamic object moving on the road while offering more accurate and dense static maps on all sequences.	-	
TLS-SLAM: Gaussian Splatting SLAM Tailored for Large-Scale Scenes	Cheng, Sicong and He, Songyang and Duan, Fuqing and An, Ning	2025	10.1109/LRA.2025.3536876	semantic features for submap creation + large-scale environments update --> extract semantic features from images using Segment Anything model --> encoding aggregated to form the memory feature vector for each submap. Usage of RGB camera + LiDAR aligment to have righ features + depth: - LiDAR pts colored based on RGB px, depth initialized using Deep Anything model reined via bilateral filtering - Semantic data from Segment Anything integrated and rendered using 3D Gaussian Splatting formulat to calculate semantic value by considering each label's contribution while accounting for others. Gaussian-based encoding to represent dynamic objs: - mean + variance Gaussian distribution for each obj based on semantic attributes, pose, and motion velocity information - parameters to weight uncertainty based on velocity components, adjusting covariance matrix to prioritize directions w/ higher velocities for accurate state estimation	feature-guided submap construction that adaptively updates submaps based on degree of feature matching between current scene and submaps, dividing complex search space into subspaces	dynamic + static objects, objects category (e.g., vehicle, pedestrian) --> E Entering (objects entering the frame), V Vanishing (objs leaving the frame), F Full, objs fully visible within frame): - if no obj px coordinates touch image boundary, object is Full - if any obj px boundary touch image boundary,, compute velocity from 3D position change to 2D camera coordinates --> vanishing are objs leaving image, otherwise Entering 3D Gaussian encoding strategy to represent 3D gaussian pts generated by dynamic objs, avoiding generationg of noisy Gaussian pts from dynamic objs that can affect model convergence Experiments showed that wo/ semantic guidance, tracking exhibits noticeable periodic drift, primarily caused by accumulated errors Combining dynamic object encoding achieves best results, where accurate rendering of mooving objs, improving visual clarity + matching ground truth	-	
DYO-SLAM: Visual Localization and Object Mapping in Dynamic Scenes	Hu, Xinggang and Wu, Yanmin and Zhao, Mingyuan and Cao, Zhenzhong andZhang, Xiangkui and Ji, Xiangyang	2025	10.1109/TCSVT.2025.3571049	Object-level map containing only static objs, represented using cubic boces and quadratic surfaces + design dynamic obj tracking algorithm and augmented reality system to demonstrate advanced applcations e.g. irtual-real occlusion and collision.	-	Semantic information acquisition and processing module: from potential moving objs based on YOLOX, perform missed detection compensation using EKG and Hungarian algorithm, used as prior knowledge for handling moving objs in localization part and as semantic labels for 2D imgs in mapping part. Localization module for dynamic scenes: - two-stage, coarse-to-fine static probability calculation for feature pts based on multiple constraints, using static probability as weight in camera pose estimation - 1. static probab of potential moving objs through motion consistency detection - 2. use DBSCAN clustering to distinguish foreground pts from background pts Object parameter estimation module: DBSCAN clustering + weighted PCA based on an energy function and minimum bounding rectangle to accratly estimate centroid, orientation, and size of objs even in dynamic scenes. Object Data Association Module: 2D, 3D and probabilistic observation information by computing association distance through color histograms, 2D IoU w/ missed detection compensation, probabilistic model based on semantic and spatial features, and 3D IoU --> Hungarian algorithm for optimal assignment to solve obj data association problem in order to enable robust and accurante association of objects observed at different times. Missed detection compensation method combining EKF and Hungarian algorithm, where EKF applied to predict state of the Bounding Box of an object, subsequently Hungarian algorithm used to associate EKF-predicted BBoxes with those detected: if predicted BBox cannot vind suitable match through Hungarian algorithm, prediction result of EKF used as BBox for the missed detection. Goal: develop dynamic obj tracking based on obj-level map constructed to have objet dynamic tracking in AR applications wo/ requiring pre-scanning object and exporting and importing 3D models. Experiments in localization comparing with ORBSLAM2,DynaSLAM, DS-SLAM, SG-SLAM, SCP-SLAM, etc: results demonstrated localization accuracy of proposed algorithm can achieve competitive results in almost all high-dynamic and low-dynamic sequences of TUM dataset. Unlike traditional AR systems that overlay virtual models on top of imgs, proposed AR system, built upon proposed approach, can realize advanced features like virtual-real occlusion, collision, and interaction, -> due to the dynamic object tracking!	-	
DGS-SLAM: Robust Visual SLAM with 3D Gaussian Splatting in Dynamic Environments	Jia, Zhaoqian and Li, Qi and Zhan, Li and Wang, Zhiguo	2025	10.1109/TCSVT.2025.3645351			DGS-SLAM: simultaneous camera tracking and dense static map construction under dynamic environments: - 3D-2D reprojection constraints through traditional feature matching - graph-based optimization - long-term data association performed by jointly considering feature matching and mask overlap relationships between frames enabling check of obj motion status within sliding window: once obj confirmed as dynamic, corresponding Gaussian components in 3DGS map along w/ related px observations in historical keyframes eliminated, preventing dynamic content from contaminating reconstructed static map - introduce dynamic-aware keyframe selection strategy based on camera movement, Gaussian co-visibility and feature matching, effectivly mitigatin risk tracking failures and map inconsistencies due to dynamic objs and rapid motion novel dynamic obj check mechanism extends obj association relationships from adjacent frames into sliding window: by validating feature point reprojection errors and monitoring global motion patterns across multiple frames, proposed method robustly differentiates truly dynamic objs from static ones wo/ relying on prior semantic categorizations. - FastInst instance segmentation to obtain px masks of objs on img - for each obj, extract ORB features within its masks -- collect feature pts from all objs in current frame + brute-force matching to find corresponding feature pts in previous frame - for texture-less objs, mask overlap relationships introduces as additional criterion "fundamental fact persists: dynamic objs inevitably disturb both SLAM and 3DGS systems" -> motion check to determine if object is indeed dynamic or not based on reprojection error. DGS-SLAM able to determine subtle movements as dynamic. Comparisons on TUM RGB-D and BONN-RGB-D dynamic datasets compared to ORB-SLAM3, DynaSLAM, GS-ICPSLAM, MonoGS, etc.: DGS-SLAM achieved lowest ATE on twelve sequences, mostly due to identification of dynamic objs thus	-	
LL-Localizer: A Lifelong Localization System Based on Dynamic i-Octree	Li, Xinyi and Yuan, Shenghai and Cai, Haoxin and Lu, Shunan and Wang, Wenhua and Liu, Jianqi	2025	10.1109/TIM.2025.3586401	Dynamic i-Octree for map management, refined bersion from original Dynamic Octree, efficiently handles randomly distributed pts: - hash pair (key,value(to define core data structure, described as Voxel (key) as the point in M (map), voxel block (value) contains data from each voxel, including map static , dynamic and prior pts - hash function construction defines voxel block index vecto based on the position / voxel scale resolution (similar to our DistanceMapSparse) - then, has index for the voxel block computed based on XOT botwise operator and multiply voxel index by large prime numbers to reduce collisions and ensure good hashing properties i-Octree: required given that prior map retained as much as possible to ensure accurate localization and map integrity --> i-Octree implementes in the voxel block - radius neighbor search finds every pt satisfying norm2 < radius --> adop pruning strategy proposed by Behley et al. with improvements to reduce computation cost + parallel computing the radius neighbor search of each pt in the scan	Load prior map first --> each pt added to prior map and static map in voxel block + construct i-Octree according to static map pts. 1. first scans -> merge them + obtain initial pose within the map using global localization module branch-and-bound Search++ (BBS++) 2. when new LiDAR scan arrives, system transforms new scan according to predicted pose --> dynamically loads local map + perform pt cloud registration to obtain accurate pose in map --> if pt in new scan fails to match static map pts, infer that environmental changes occurred in that region or prior maps lacking --> update this new pt failed association to temporary map for future map updates; conversely, if pt successfully finds suitable map pts in static or prior map, no update made! 3. once size of temporary map in voxelblock reaches predefined threshold, static map updated + i-Octree updated accordingly HDL-Loc fails when system encounters areas wo/ prior map due to its inability to update maps. Proposed system effectively handles both scenarios by dynamically loading local maps + applying three-layer map management approach + simultaneously update maps of areas where enviroment has changed and areas wo/ prior	-		
MSSD-SLAM: Multifeature Semantic RGB-D Inertial SLAM With Structural Regularity for Dynamic Environments	Wang, Yanan and Tian, Yaobin and Chen, Jiawei and Chen, Cheng and Xu, Kun and Ding, Xilun	2025	10.1109/TIM.2024.3509541	-	-	MSSD-SLAM: novel semantic visual-inertial SLAM method that integrated pt, line, plane features and combine IMU and visual data --> incorporate multiple sources of information into multistage Dynamic Filter (filtering out dynamic features). Dynamic filter uses semantic and structure regularity of environment (Manhattan world assumption) + state information from IMU integration --> to compute dynamic probabilities of landmarks. - forward and backward projections among landmarks of local keyframes to further discern dynamic landmarks - during BA optimization of local mapping, landmarks assigned ifferent weights according to their dynamic probabilities - compute long-term dynamic object probabilities by projecting local keyframes onto one another, derived dynamic probabilities from multiple sources merged to form dynamic probabilities associated with corresponding landmarks where any landmark w/ dynamic probability above threhold is flagged as outlier, whereas those falling below threshold retained for pose estimation and mapping! Experiments conducts on TUM RGB-D and self-recorded dataset. algorithm based on ORB-SLAM3 compared with other ones (DS-SLAM, SG-SLAM, Dynamic-VINS, and DRG-SLAM). Proposed method outperformed other methods on TUM RGB-D dynamic dataset, w/ average RMSE improvements up to 92% compared w/ original ORB-SLAM3. Impact of removing structural regularity or project led to notable 35% and 46% accuracy enhancement over using	-	
SPLIN: A Structured Plane-Based LiDAR-Inertial SLAM With Dynamic Object Removal	Zhu, Jun and Mi, Weiming and Li, Hongyi and Xue, Tao and Zhao, Jiawei and Wang, Shengjie and Zhang, Tao	2025	10.1109/TIM.2025.3643083	-	-	SPLIN (Structured Plane-based Lidar-Inertial) framework with dynamic object removal (DOR) and uncertainty-aware optimization: - incremental static-referenced DOR (ISDOR) that exploits stable geometric signatures of static structures to incrementally construct voxelized static map --> coarse-to-fine strategy flags dynamic pts upon their first observation at frame level ISDOR: leveragin static structures especially ground, classify dynamic pts into 1. pts lying above static regions + 2. vertically "suspended" pts w/ significant height + 3. pts adjacent to areas previous flagged as dynamic --> problem is catching first time pts appear dynamic. - constructu dynamically updated 3-D voxel map that stores spatial idistribution of previously observed static pts and adopt corase-to-fine strategy to detect dynamic objs - once portion of dynamic ojs identified w/ high confidence, complete obj can be recovered via region growing and clustering, avoiding need for groudn-plane fitting or projection onto depth img In experiments, SDOR produces segmentations that visually align with the GT, accurately isolating dynamic points while preserving static structure. Despite adaptive downsampling, highly confined geometries can induce anisotropic observability with insufficient directional excitation, which weakens geometric constraints. In addition, the current ISDOR is less effective for low-profile or quasi-static movers, and its performance deteriorates in scenes exhibiting rapid	-	
Estimating the dominant frequencies of real-valued cyclic processes in natural environments for long-term operation of autonomous robots	Vintr, Tomas and Blaha, Jan and Ulrich, Jiří and Rouček, Tomas and Broughton, George and Duckett, Tom and Arvin, Farshad and Krajník, Tomáš	2026	10.1016/j.robot.2025.105299	-	-	-	Maps that explicity model environment dynamics called Maps of Dynamics (MoD) ~ queryable models of spatial or spatio-temporal patterns of dynamics. Propose apprach in which the most crucial part of Non-Uniform Fourier Transform (NUFT)-based Temporal MoDs does not depend on a manually pre-selected set of candidate frequencies, where dominant frequencies are inferred directly from data using iterative sampling-based search for frequency dominant w.r.t. other frequencies, giving abilityy to lear temporal pattersn wo/ relying on prior knowledge or assumptions about its environment. Method showed that proposed method needs around 4 days long dataset of a phenomenon, whose periores derived primarily from one/day and one/week periods, to provide predictions one week into future comparable to those of the original method that includes a set of candidates of angular frequencies + method needed around 3 weeks long dataset to stabilise and not show negative deviations -->> predictions of both methods for one week into future were comparable on an almost one-year-long dataset, however in situations of significant changes in number of detections of phenomenon, error of proposed method compared to original was raised. When experiments whose phenomenon not derived from calendaar, proposed method created incomparably better model than original approach and provided satisfactory predictions. Showed that the proposed improvement of FreMEn increased the variety of environments where an autonomous robot can successfully operate without prior knowledge of the underlying processes and solve a wide variety of tasks similar to those tested in the human environments.	